4. Distributions

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Outline

1. Bernoulli and Binomial Distributions
2. Hypergeometric Distribution
3. Geometric and Negative Binomial Distributions
4. Poisson Distribution
5. Uniform, Exponential, and Friends
6. Other Continuous Distributions
7. Normal Distribution: Basics
8. Standard Normal Distribution
9. Sample Mean of Normals
10. The Central Limit Theorem + Proof
11. Central Limit Theorem Examples
12. Extensions — Multivariate Normal Distribution
13. Extensions — Lognormal Distribution
14. Computer Stuff

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In the next few lessons we'll discuss some important **discrete** distributions:

- Bernoulli and Binomial Distributions
- Hypergeometric Distribution
- Geometric and Negative Binomial Distributions
- Poisson Distribution

Definition: The **Bernoulli distribution** with parameter p is given by

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Definition: The **Binomial distribution** with parameters n and p is given by

$$P(Y = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \ldots, n.$$

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Think of the Binomial as the number of successes from n Bern(p) trials.

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$$P(\text{exactly 3 reds are picked}) \;=\; \frac{\binom{15}{3}\binom{10}{4}}{\binom{25}{7}}. \quad \square$$

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So same mean as the Hypergeometric, but slightly larger variance.

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We'll get the mean and variance of the Geometric via the mgf. . . .

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Remark: We could also have proven this directly from the definition of expected value.

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Answer: $\mathrm{E}[Z] = 1/p = 6$ tosses. $\square$

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Why is it called the Memoryless Property? If an event hasn't occurred by time s, the probability that it will occur after an additional t time units is the same as the (unconditional) probability that it will occur after time t — it forgot that it made it past time s!

Proof: First of all, for any $t = 0, 1, 2, \ldots$, the tail probability is

$$P(Z > t) \;=\; P(t \text{ Bern}(p) \text{ failures in a row}) \;=\; q^t. \qquad (*)$$

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$$\begin{aligned} P(Z > s + t | Z > s) &= \frac{P(Z > s + t \cap Z > s)}{P(Z > s)} \\ &= \frac{P(Z > s + t)}{P(Z > s)} \\ &= \frac{q^{s+t}}{q^s} \quad \text{(by } (*)\text{)} \\ &= q^t \\ &= P(Z > t). \quad \square \end{aligned}$$

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Fun Fact: The Geom(p) is the only discrete distribution with the memoryless property.

Not-as-Fun Fact: Some books define the Geom(p) as the number of Bern(p) *failures* until you observe a success. $\#$ failures $= \#$ trials $- 1$. You should be aware of this inconsistency, but don't worry about it now.

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Remark: As with the Geom(p), the exact definition of the NegBin depends on what book you're reading.

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1. Bernoulli and Binomial Distributions
2. Hypergeometric Distribution
3. Geometric and Negative Binomial Distributions
4. Poisson Distribution
5. Uniform, Exponential, and Friends
6. Other Continuous Distributions
7. Normal Distribution: Basics
8. Standard Normal Distribution
9. Sample Mean of Normals
10. The Central Limit Theorem + Proof
11. Central Limit Theorem Examples
12. Extensions — Multivariate Normal Distribution
13. Extensions — Lognormal Distribution
14. Computer Stuff

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Anti-Example: Customers arrive at a restaurant. They show up in groups, not one-at-a-time. The rate varies over the day (more at dinnertime). Arrivals may not be independent. This ain't a Poisson process. $\square$

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To begin with, let's define $P_x(t) \equiv P(N(t) = x)$, i.e., the probability of exactly x arrivals by time t.

Definition: Let X be the number of occurrences in a Poisson(λ) process in a *unit interval* of time. Then X has the **Poisson distribution** with parameter λ.

Notation: $X \sim \text{Pois}(\lambda)$.

Theorem/Definition: $X \sim \text{Pois}(\lambda) \Rightarrow P(X = k) = e^{-\lambda}\lambda^k/k!$, $k = 0, 1, 2, \ldots$.

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Note that the probability that there haven't been any arrivals by time $t + h$ can be written in terms of the probability that there haven't been any arrivals by time t. . . .

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Thus, $\mathrm{Var}(X) = \mathrm{E}[X^2] - (\mathrm{E}[X])^2 = \lambda(1+\lambda) - \lambda^2 = \lambda.$ $\square$

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1. Bernoulli and Binomial Distributions
2. Hypergeometric Distribution
3. Geometric and Negative Binomial Distributions
4. Poisson Distribution
5. Uniform, Exponential, and Friends
6. Other Continuous Distributions
7. Normal Distribution: Basics
8. Standard Normal Distribution
9. Sample Mean of Normals
10. The Central Limit Theorem + Proof
11. Central Limit Theorem Examples
12. Extensions — Multivariate Normal Distribution
13. Extensions — Lognormal Distribution
14. Computer Stuff

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Remark: Look at $\mathrm{E}[X]$ and $\mathrm{Var}(X)$ for the Geometric distribution and see how they're similar to those for the exponential. (Not a coincidence.)

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Example: If $X \sim \text{Exp}(\lambda)$, then $S(t) = \lambda e^{-\lambda t}/e^{-\lambda t} = \lambda$. So if X is the exponential lifetime of a lightbulb, then its instantaneous burn-out rate is always λ — always good as new! This is clearly a result of the Memoryless Property. $\square$

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Theorem: Amazingly, it can be shown (after a lot of work) that the interarrival times of a PP are *all* iid Exp(λ)! See for yourself when you take a stochastic processes course.

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$$P(X > 4 \text{ min}) \;=\; P(X > 1/15 \text{ hr}) \;=\; e^{-\lambda t} \;=\; e^{-20/15}. \quad \square$$

Definition/Theorem: Suppose $X_1, \ldots, X_k \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$, and let $S = \sum_{i=1}^{k} X_i$. Then S has the **Erlang distribution** with parameters k and λ, denoted $S \sim \text{Erlang}_k(\lambda)$.

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Notice that the cdf is the sum of a bunch of Poisson probabilities. (Won't do it here, but this observation helps in the derivation of the cdf.)

Expected value, variance, and mgf:

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$$P(X+Y<1) = 1 - \sum_{i=0}^{k-1} \frac{e^{-\lambda s}(\lambda s)^i}{i!} = 1 - \sum_{i=0}^{2-1} \frac{e^{-(2\cdot 1)}(2\cdot 1)^i}{i!} = 0.594. \quad \square$$

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Remark: The Gamma distribution generalizes the Erlang distribution (where α has to be a positive integer). It has the same expected value and variance as the Erlang, with α in place of k.

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Remark: The Gamma distribution generalizes the Erlang distribution (where α has to be a positive integer). It has the same expected value and variance as the Erlang, with α in place of k.

Remark: If α is a positive integer, then $\Gamma(\alpha) = (\alpha - 1)!$. Party trick: $\Gamma(1/2) = \sqrt{\pi}$.

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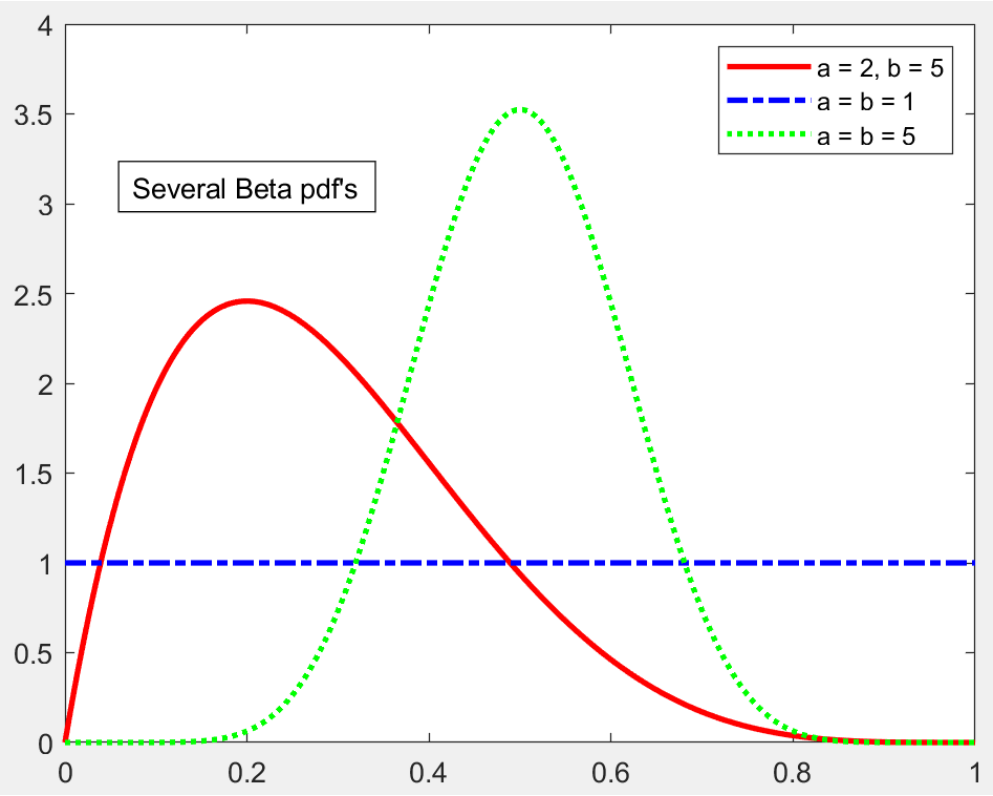

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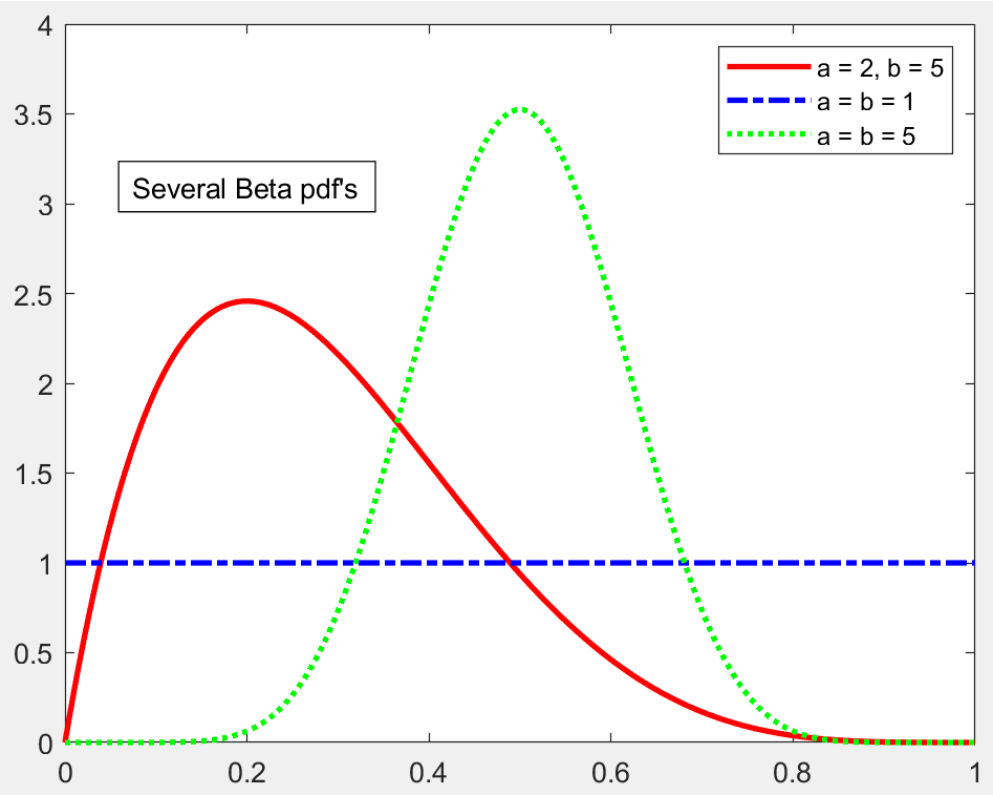

Remark: Certain versions of the Uniform ($a = b = 1$) and Triangular distributions are special cases of the Beta.

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The probability that it fails before 2000 hrs is

$$F(2000) \;=\; 1 - \exp[-(2000/200)^{1/3}] \;=\; 0.884. \quad \square$$

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Weird Fact: $X_1, \ldots, X_n \overset{\text{iid}}{\sim}$ Cauchy $\Rightarrow \sum_{i=1}^n X_i/n \sim$ Cauchy. Even if you take the average of a bunch of Cauchys, you're right back where you started!

Alphabet Soup of Other Distributions

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Examples: Heights, weights, SAT scores, crop yields, and averages of things tend to be normal.

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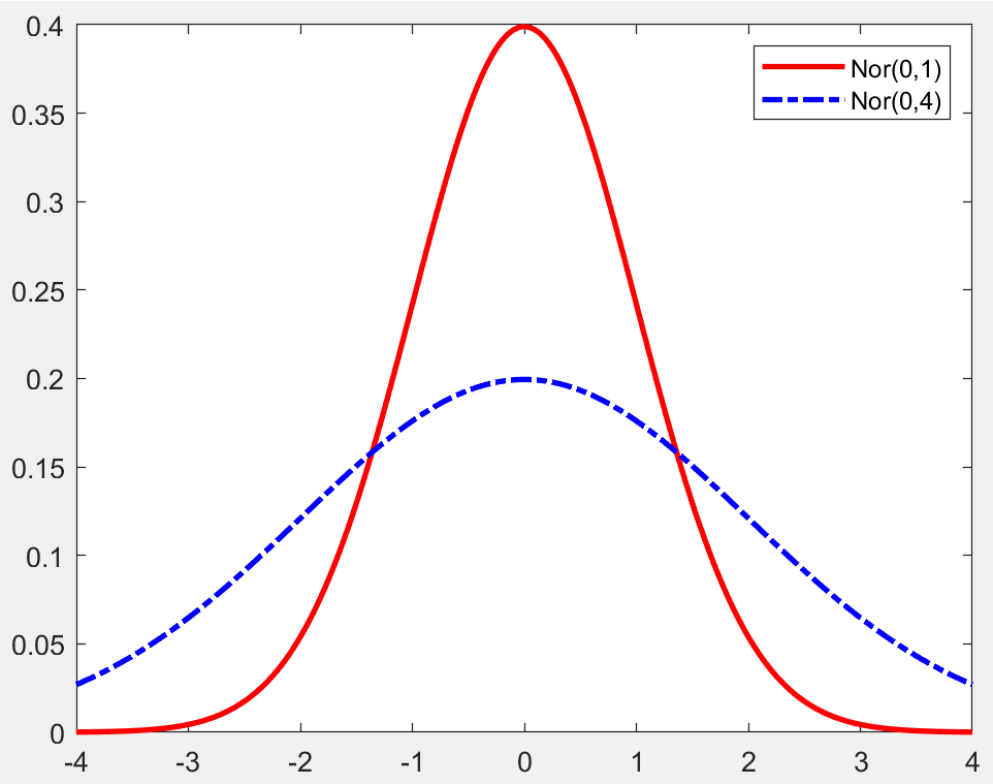

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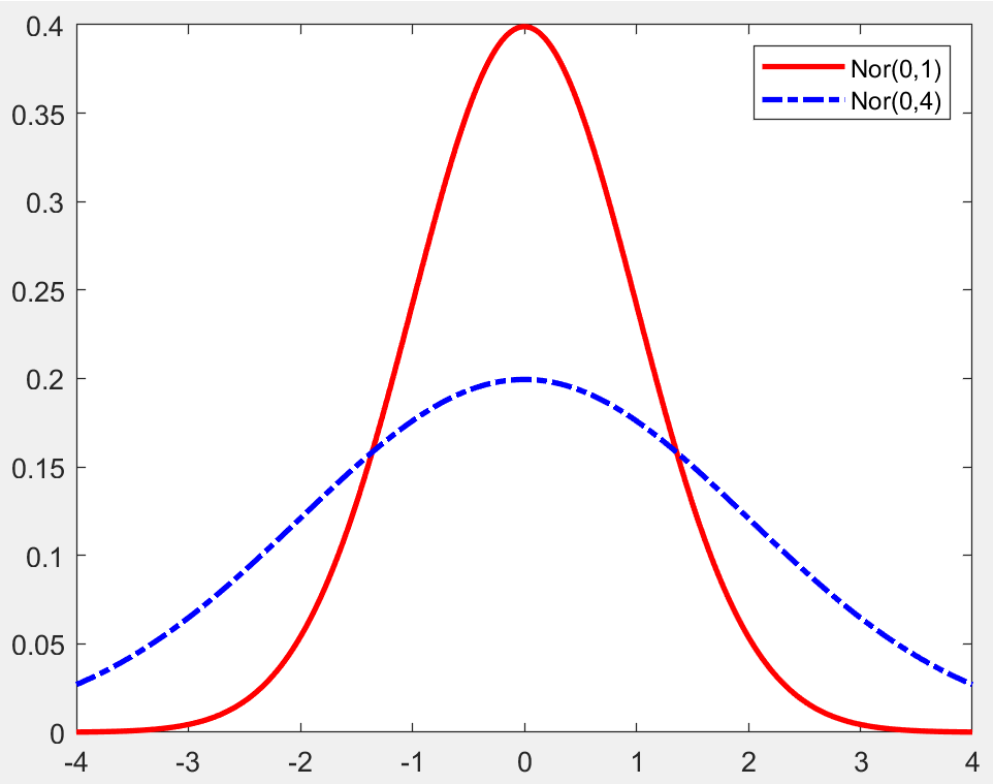

Small σ^2 corresponds to a "tall, skinny" bell curve; large σ^2 gives a "short, fat" bell curve.

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Proof: Integration by parts or mgf (below).

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Proof: Calculus (or look it up in a table of integrals).

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and we are done by mgf uniqueness. $\square$

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$$\text{Var}(2X - 3Y) \;=\; 4\text{Var}(X) + 9\text{Var}(Y) \;=\; 70.$$

Remark: A normal distribution is *completely characterized* by its mean and variance.

By the above, we know that a linear combination of independent normals is still normal. Therefore, when we add up independent normals, all we have to do is figure out the mean and variance — the normality of the sum comes for free.

Example: $X \sim \text{Nor}(3, 4)$, $Y \sim \text{Nor}(4, 6)$, and X, Y are independent. Find the distribution of $2X - 3Y$.

Solution: This is *normal* with

$$\text{E}[2X - 3Y] \;=\; 2\text{E}[X] - 3\text{E}[Y] \;=\; 2(3) - 3(4) \;=\; -6$$

and

$$\text{Var}(2X - 3Y) \;=\; 4\text{Var}(X) + 9\text{Var}(Y) \;=\; 70.$$

Thus, $2X - 3Y \sim \text{Nor}(-6, 70)$. $\square$

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The manipulation described in this corollary is referred to as **standardization**.

1. Bernoulli and Binomial Distributions
2. Hypergeometric Distribution
3. Geometric and Negative Binomial Distributions
4. Poisson Distribution
5. Uniform, Exponential, and Friends
6. Other Continuous Distributions
7. Normal Distribution: Basics
8. Standard Normal Distribution
9. Sample Mean of Normals
10. The Central Limit Theorem + Proof
11. Central Limit Theorem Examples
12. Extensions — Multivariate Normal Distribution
13. Extensions — Lognormal Distribution
14. Computer Stuff

Lesson 4.8 — Standard Normal Distribution

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So the probability that *any* normal RV is within k standard deviations of its mean doesn't depend on the mean or variance.

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Finally, note that Toyota's six-sigma corresponds to $2\Phi(6) - 1 \doteq 1.0000$.

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&= \Phi(0.75) - [1 - \Phi(1)] \\
&= 0.7734 - [1 - 0.8413] \;=\; 0.6147. \quad \square
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$$\begin{aligned} W - M &\sim \text{Nor}(\text{E}[W - M], \text{Var}(W - M)) \\ &\sim \text{Nor}(65 - 68, 1 + 4) \sim \text{Nor}(-3, 5). \end{aligned}$$

Then

$$\begin{aligned} P(W > M) &= P(W - M > 0) \\ &= P\left(Z > \frac{0 + 3}{\sqrt{5}}\right) \\ &= 1 - \Phi(3/\sqrt{5}) \\ &\doteq 1 - 0.910 = 0.090. \quad \square \end{aligned}$$

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Lesson 4.9 — Sample Mean of Normals

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In the upcoming statistics portion of the course, we'll learn that this makes $\bar{X}$ an excellent **estimator** for the mean μ, which is typically unknown in practice.

Example: Suppose that $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Nor}(\mu, 16)$. Find the sample size n such that

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So if you take the average of 62 observations, then $\bar{X}$ has a 95% chance of being within 1 of μ. $\square$

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where " $\xrightarrow{d}$ " means that the cdf of $Z_n \to$ the $\mathrm{Nor}(0,1)$ cdf.

Slightly Honors Informal Proof: Suppose that the mgf $M_X(t)$ of the X_i's exists and satisfies certain technical conditions that you don't need to know about. (OK, $M_X(t)$ has to exist around $t = 0$, among other things.)

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where the last equality is from our standardization assumption.

So after all of this build-up, we have

$$\lim_{y \to 0} \frac{\ell\mathrm{n}\big(M_X(ty)\big)}{y^2}$$

$$= \lim_{y \to 0} \frac{t\, M_X'(ty)}{2y\, M_X(ty)} \quad \text{(by (3) et L'Hôspital to deal with } 0/0\text{)}$$

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(5) In fact, there are versions of the CLT that are actually a lot more general than the theorem presented here!

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Lesson 4.11 — Central Limit Theorem Examples

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Example: To show that the Central Limit Theory really works, let's add up just $n = 12$ iid Unif(0,1)'s, $U_1, \ldots, U_n$. Let $S_n = \sum_{i=1}^{n} U_i$.

Lesson 4.11 — Central Limit Theorem Examples

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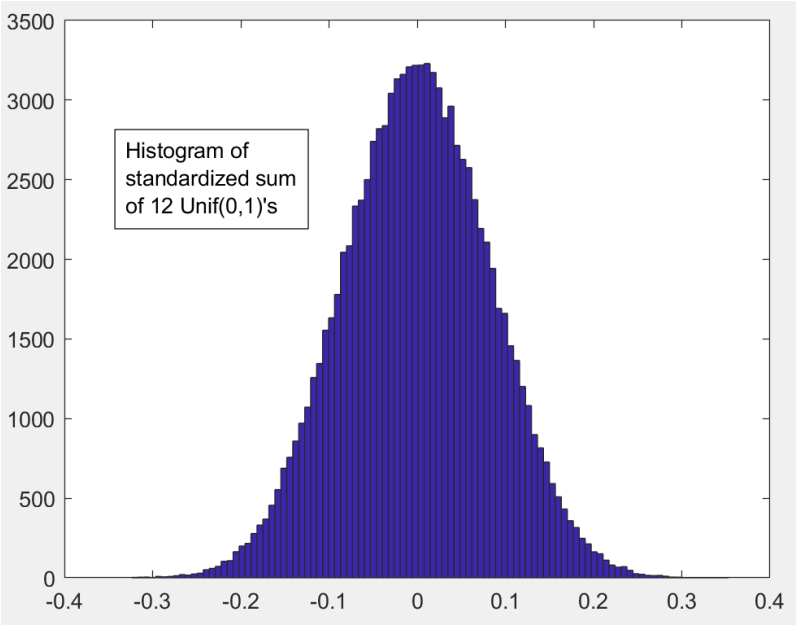
The histogram was compiled using 100,000 simulations of Z_{12}. It works! □



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For our problem, $\lambda = 1/1000$ and $n = 100$, so that $\text{E}[\bar{X}] = 1000$ and $\text{Var}(\bar{X}) = 10000$.

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Remark: This problem can be solved exactly if we have access to the Excel Erlang cdf function `GAMMADIST`. And what do you know, you end up with exactly the same answer of 0.383!

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Notice that we didn't care whether or not the data came from an exponential distribution. We just needed the mean and variance. $\square$

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$$\frac{Y - \text{E}[Y]}{\sqrt{\text{Var}(Y)}} = \frac{Y - np}{\sqrt{npq}} \approx \text{Nor}(0, 1).$$

The usual rule of thumb for the Normal approximation to the Binomial is that it works pretty well as long as $np \geq 5$ and $nq \geq 5$.

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Example: Suppose $Y \sim \text{Bin}(100, 0.8)$ and we want

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Note that it incorporates a **"continuity correction"** to account for the fact that the Binomial is *discrete* while the Normal is *continuous*. If you don't want to use it, don't worry too much.

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The actual answer (using the true Bin(100,0.8) distribution) turns out to be 0.1923 — pretty close! $\square$

1. Bernoulli and Binomial Distributions
2. Hypergeometric Distribution
3. Geometric and Negative Binomial Distributions
4. Poisson Distribution
5. Uniform, Exponential, and Friends
6. Other Continuous Distributions
7. Normal Distribution: Basics
8. Standard Normal Distribution
9. Sample Mean of Normals
10. The Central Limit Theorem + Proof
11. Central Limit Theorem Examples
12. Extensions — Multivariate Normal Distribution
13. Extensions — Lognormal Distribution
14. Computer Stuff

Lesson 4.12 — Extensions — Multivariate Normal Distribution

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Definition: (X, Y) has the **Bivariate Normal distribution** if it has pdf

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Pretty nasty joint pdf, eh?

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This guy doesn't have much chance of having a good GPA. $\square$

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where μ is related to the "drift" of the stock price (i.e., the natural rate of increase), σ is its "volatility" (how much the stock bounces around), $S(0)$ is the initial price, and Z is a standard normal RV.

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- If IBM is selling for \$95, then I won't exercise my option, and will walk away with my tail between my legs having lost my original \$3.

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Black and Scholes won a Nobel Prize for calculating $\mathrm{E}[C]$. We'll get the same answer via a different method — but, alas, no Nobel Prize. ☹

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There are many generalizations of this problem that are used in practical finance problems, but this is the starting point. Meanwhile, get your tickets to Norway or Sweden or wherever they give out the Nobel! ☺

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Lesson 4.14 — Computer Stuff

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Evaluating pmf's / pdf's and cdf's

Lesson 4.14 — Computer Stuff

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Functions such as `NORMSINV` and `TINV` can calculate the inverses of the standard normal and t distributions, respectively.

Simulating Random Variables

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Motivation: Simulations are used to evaluate a variety of real-world processes that contain inherent randomness, e.g., queueing systems, inventory systems, manufacturing systems, etc. In order to run simulations, you need to generate various RVs, e.g., arrival times, service times, failure times, etc.

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- It can be shown that $X = \lceil \ell\text{n}(U)/\ell\text{n}(1-p) \rceil \sim \text{Geom}(p)$, where $\lceil \cdot \rceil$ is the "ceiling" (integer round-up) function.

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- To get $X \sim \text{Nor}(\mu, \sigma^2)$ from $Z \sim \text{Nor}(0, 1)$, just take $X = \mu + \sigma Z$.
- Amazingly, it's "Muller", not "Müller".
  See https://www.youtube.com/watch?v=nntGTK2Fhb0

Honors Proof: We follow the method from Module 3 to calculate the joint pdf of a function of two RVs. Namely, if we can express $U_1 = k_1(Z_1, Z_2)$ and $U_2 = k_2(Z_1, Z_2)$ for some functions $k_1(\cdot, \cdot)$ and $k_2(\cdot, \cdot)$, then the joint pdf of (Z_1, Z_2) is given by

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so that

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where we (non-rigorously) get rid of the "$\pm$" to balance off the fact that the range of $y = \arccos(x)$ is only regarded to be $0 \le y \le \pi$ (not $0 \le y \le 2\pi$).

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\end{aligned}
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which is the product of two iid Nor(0,1) pdf's, so we are done! ☺